Finite coset geometries in Highly symmetric hypertopes

Human Compression project

Abstract

We verify the three finite examples and all twenty-three table entries in Fernandes–Leemans–Weiss, using their exact markings and presentations. The common coset criteria are acquired from the earlier recovery of *Hexagonal extensions of toroidal hypermaps*; their application here includes the actual flag and residue tests. All printed orders and regularity assertions hold. One group name and both chiral flag-count captions require correction. The completeness conjecture concerning the chiral list is not used or proved.

1 The groups and the certificate interface

We use the accepted author manuscript of the source [1], with physical page numbers. Its Table 2, on page 20, specifies the presentations

$$P(p,a,b) = \langle x,y,z \mid x^3, y^3, z^p, (x^{-1}z)^2, (y^{-1}z)^2, (x^{-1}y)^3, (xy^{-1}x)^a(xy)^b \rangle. \tag{1}$$

The involutory completion, when it exists, is part of the conclusion and is not imposed in this presentation. Put $a_0 = 1, a_1 = x, a_2 = y, a_3 = z$ and

$$P_J = \langle a_i^{-1}a_j : i,j \in J \rangle, \qquad P_i = P_{\{0,1,2,3\}\setminus\{i\}}.$$

The associated incidence system has right cosets of P_i as its elements. Two elements of different types are incident when their cosets intersect.

For Table 1 we use standard polytope rotations u, v, w. The literal source relation (5.1) gives

$$Q(p,a,b) = \langle u,v,w \mid u^6, v^3, w^p, (uv)^2, (vw)^2, (uvw)^2, (vu^{-2})^a(v^{-1}u^2)^b \rangle. \tag{2}$$

Indeed $u = \rho_0\rho_1$, $v = \rho_1\rho_2$, $w = \rho_2\rho_3$, so $vu^{-2} = \rho_1\rho_2(\rho_1\rho_0)^2$ and $v^{-1}u^2 = \rho_2\rho_1(\rho_0\rho_1)^2$. The pair-difference marking for this group is $(1, u, uv, uvw)$. In the program the three nonidentity entries are the free generators; this is the invertible change $u = X, v = X^{-1}Y, w = Y^{-1}Z$. It is essential to keep this change of marking when checking inversion.

We acquire the following complete finite certificate theorem from [2]. For four maximal rotation parabolics, set $K_J = \bigcap_{j\in J} P_j$. The type-$i$ completions of the base flag of types J are counted by

$$c(i,J) = \#\{g \in T : g \in P_iP_j \ (j \in J)\},$$

where T is a right transversal of $P_i \cap P_{j_0}$ in P_{j_0} for one $j_0 \in J$. We check full pair-difference intersection, the equalities $c(i,J) = [K_J : K_J \cap P_i]$ for $|J| = 2$, and $c(i,J) = 2$ for $|J| = 3$. In addition, the P_i generate P, each P_f is generated by its intersections with the other parabolics, and every rank-two residue is connected. The last test uses its actual incidence graph: two copies of K_J, with h joined to k exactly when $hk^{-1} \in P_iP_j$, for the two remaining types i, j.

These conditions imply a thin residually connected geometry with a free action having exactly two chamber orbits. Thus they check more than marked subgroup intersections. Simultaneous inversion of the three marked generators is tested by evaluating *all* defining relators on their inverses. If the relators hold, the induced endomorphism squares to the identity and is an automorphism. Otherwise a failed relator is a literal obstruction. The acquired two-orbit theorem [3] then determines the full type-preserving automorphism group. In the regular case we also construct $H = P \rtimes \langle t \rangle$, put $\rho_i = ta_i$, and verify all 256 subgroup intersections and all normalized flag-completion tests in H. The acquired regular-completion proof identifies this geometry with the rotation coset geometry.

2 All table entries

Theorem 1. *The presentations* (1) *and* (2)*, at precisely the parameters in Table 1, give the regular or chiral geometries recorded there. The displayed numbers are the orders of their full type-preserving automorphism groups. Each regular geometry has that many chambers; each chiral geometry has twice that many chambers.*

p	(a,b)	Table 1 order	Table 2 order	status
3	(2,0)	240	120	regular
3	(3,0)	1296	648	regular
3	(4,0)	15360	7680	regular
3	(1,2)	336	168	chiral
3	(1,3)	2184	1092	chiral
3	(1,4)	8064	4032	chiral
3	(2,2)	2880	1440	regular
3	(2,3)	6840	3420	chiral
4	(1,1)	288	—	regular
4	(1,2)	2016	1008	chiral
4	(2,0)	768	384	regular
5	(2,0)	28800	14400	regular

Table 1: The exact finite parameters.

Proof. For every row, GAP computes an isomorphism from the finitely presented group to a finite permutation group. This supplies both an upper bound for the presentation and a faithful finite image; an arbitrary finite quotient would not suffice. The complete seven-word relator lists are retained. A separate literal-permutation checker verifies the relators, generating orders and all pair-difference subgroup intersections without calling the presentation routine.

The acquired finite criteria of Section 1 are then checked for every row. There are 256 pair-difference intersections, sixteen completion counts, four parabolic-generation tests and six actual rank-two residue graphs per row. The absent inversion is witnessed by the last relator in each chiral row. Each regular row has the full involutory completion and its independent intersection and flag checks. The proper-flag counts prove flag completion and transitivity on all proper types; the two completions of a panel lie in different free chamber orbits. Consequently the full groups and chamber counts are those asserted. For (2) we verify the polytope assertion directly. For each $i < j < k$, the literal subgroups satisfy $P_j \subseteq P_iP_k$. To use this condition, normalize a middle face to P_j. Every incident face of type i or k has a representative in P_j, so write the two faces as P_ih and P_kt with $h, t \in P_j$. Since $ht^{-1} \in P_j \subseteq P_iP_k$, the two faces are incident. Thus incidence, ordered by type, is transitive. It is antisymmetric, and adjoining a least and greatest face gives a ranked partially ordered set. The verified geometry property gives its full-length maximal chains; thinness gives the diamond condition, and residue connectivity gives strong flag connectivity. These are exactly the abstract polytope axioms. The forty-eight subgroup-product inclusions are independently checked by literal set products. No string reconstruction theorem is needed for these finite rows.

Local residue identifications are marked isomorphisms. In (1), $\langle x, y\rangle$ is the exact torus group

$$\langle x, y \mid x^3, y^3, (x^{-1}y)^3, (xy^{-1}x)^a(xy)^b\rangle$$

of order $3(a^2 + ab + b^2)$. Its presentation maps onto the actual parabolic, and independently computed equal orders prove injectivity. The two designated spherical parabolics are checked in exactly the same way against $\langle s, t \mid s^3, t^p, (s^{-1}t)^2\rangle$. For (2) the facet group has presentation $\langle u, v \mid u^6, v^3, (uv)^2, (vu^{-2})^a(v^{-1}u^2)^b\rangle$, of order $6(a^2 + ab + b^2)$; the vertex group is the spherical

$\langle v, w \mid v^3, w^p, (vw)^2 \rangle$. The actual coset criteria already prove these groups describe residues; the local names are not inferred solely from an unmarked order match. □

The statement about chambers corrects the captions of both original tables. For example, Table 2 at $(p, a, b) = (3, 1, 2)$ has full automorphism group of order 168 and 336 chambers. A "flag" in the table caption must mean a chamber; even under that interpretation the printed g is not the number of flags in a chiral row.

3 The relation between the tables and the group names

The group of each Table 2 row embeds with index two in the corresponding Table 1 full group. This assertion is checked as a marked map, not by comparing the two orders. In the rotations (u, v, w) of (2), the map from (1) is

$$x \longmapsto v^{-1}u^{-2}, \qquad y \longmapsto v^{-1}, \qquad z \longmapsto w. \tag{3}$$

Every relator is satisfied, the image has half the order of Q, and the computed homomorphism is bijective onto that subgroup. Interchanging the first two images changes the orientation of the torus parameter; we retain (3) literally, including in the chiral rows. In the regular completions the corresponding involution map is

$$(r_0, r_1, r_2, r_3) \longmapsto (\rho_2, \rho_1^{\rho_0}, \rho_1, \rho_3).$$

The omitted generator ρ_0 is an involution outside this index-two subgroup. In the chiral cases an explicit involution outside the image is likewise retained. These data verify the split doubling assertions, including the unspecified actions in the source's semidirect-product notation. No converse classification is inferred from this construction.

The symmetric, projective linear, direct-product and real Coxeter names are checked by actual isomorphisms to their standard finite models. The real Coxeter models use their complete reflection presentations. The order-648 group is the monomial group $G(3, 3, 4)$:

$$\{(e_1, e_2, e_3, e_4) \in \mathbb{F}_3^4 : \sum e_i = 0\} \rtimes S_4,$$

acting on four blocks of three points. An explicit isomorphism to this model is retained. For the order-7680 group we use the named CHEVIE model G_{29} [4]. Its four roots are the rows of

$$\frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & -2 \\ -i & i & 1 & 1 \\ 1-i & -1+i & 0 & 0 \\ 0 & 1-i & -1+i & 0 \end{pmatrix}.$$

The reflections $v \mapsto v - 2\langle v, r\rangle r/\langle r, r\rangle$ act on the orbit of these roots. Exact arithmetic in $\mathbb{Q}(i)$ gives 160 roots and a permutation group of order 7680. Since the four roots span, that permutation action is faithful. The actual isomorphism to the Table 2 group is checked. These explicit models remove any reliance on the order alone for the source's abbreviations $[1\,1\,2]^3$ and $[1\,1\,2]^4$. The historical abbreviations are retained as labels; the displayed models specify the groups used here.

For $(p, a, b) = (3, 1, 4)$ the characteristic perfect residual has order 336 and is explicitly isomorphic to $\mathrm{SL}_2(7)$. We retain a subgroup A of order twelve, its isomorphism to A_4, and the equalities $A \cap N = 1$, $NA = G$. Thus the source's $\mathrm{SL}_2(7) \rtimes A_4$ is proved with its actual conjugation action. Its further extension in Table 1 is the split doubling above.

Proposition 2. *The group in Table 2 with $(p, a, b) = (3, 2, 2)$ is*

$$D = \{(s, t) \in S_5 \times S_4 : \mathrm{sgn}(s) = \mathrm{sgn}(t)\},$$

not the printed $A_5 \times S_4$.

Proof. On disjoint sets of five and four points, D is generated by

$$(1\,2\,3),\ (3\,4\,5),\ (6\,7\,8),\ (7\,8\,9),\ (1\,2)(6\,7).$$

These generate $A_5 \times A_4$ and one element of odd sign on each factor, hence the entire parity subgroup, of order 1440. The certificate gives a bijective homomorphism from the presented Table 2 group to this model.

The stable last term of the derived series of D is $A_5 \times 1$. Indeed the first terms are $A_5 \times A_4$, $A_5 \times V_4$ and $A_5 \times 1$. Its centralizer in D is $1 \times A_4$, of order twelve, since $C_{S_5}(A_5) = 1$. In contrast, the stable derived subgroup of $A_5 \times S_4$ is also $A_5 \times 1$, but its centralizer has order 24. This characteristic-subgroup invariant excludes an isomorphism. The finite checker verifies both derived series and centralizer orders. □

4 The three finite examples

Proposition 3. *Examples 4.4, 4.5 and 6.3 have all the group and incidence properties asserted in the source.*

Proof. For Example 4.4 put $u = ab$, $v = bc$, $t = b$, where a, b, c are the three printed involutions. Then t inverts u and v, and $(abc)^2 = 1$ is precisely $[u, v^{-1}] = 1$. Thus the presentation is $(C_3 \times C_3) \rtimes C_2$, with simultaneous inversion, of order 18. Conversely this model satisfies every defining relation, so it also supplies the lower bound. The three maximal parabolics have order six, and any two intersect in their common involution. They therefore have product of size 18. Every pair of cosets of different types is incident, giving $K_{3,3,3}$. All flags extend; there are 27 chambers, every panel has three completions, and every residue of rank at least two is connected. Its full type-preserving group is S_3^3. In particular the designated order-18 group is not flag transitive.

For Example 4.5 use the literal generators

$$(12)(34), \quad (26)(35), \quad (14)(23), \quad (14)(35).$$

They generate A_6 and satisfy all 256 marked subgroup intersections. Their maximal parabolics have orders $(24, 60, 60, 24)$ and the four type sets have sizes $(15, 6, 6, 15)$. The complete coset calculation gives 381 incidence edges, 1080 three-type flags and 720 chambers. All shorter flags extend. Among the 1080 panels, 720 have two completions and 360 have four. A residue of types $1, 2$ through a base flag of types $0, 3$ is $K_{4,4}$. A base flag of types $1, 2$ has a residue consisting of two disjoint 10-cycles. There are precisely 36 disconnected residues of rank at least two. Hence thinness, residual connectivity and designated flag transitivity all fail. The last assertion also follows from $|A_6| = 360 < 720$.

For Example 6.3 retain exactly the four permutations printed on page 14. Their group has order 72, with marked subgroup orders, in binary subset order,

$$1, 2, 2, 6, 2, 4, 8, 24, 2, 6, 6, 18, 4, 12, 24, 72.$$

All 256 intersections are correct. The rotation subgroup is the whole group. For $I = \{0, 1, 2\}$ and $J = \{1, 2, 3\}$,

$$|P_I| = |P_J| = 24, \qquad |P_I \cap P_J| = 8, \qquad |P_{I \cap J}| = 4.$$

The source's second involution ρ_1 belongs to the first intersection and not to $P_{I \cap J}$, giving a literal witness. These computations are independently repeated by closure of the permutation arrays and set intersections, without subgroup or coset routines from GAP. For Examples 4.4 and 4.5 this independent check also constructs every coset, every flag and every residue incidence graph. □

5 Scope, parameters and attribution

A minor domain qualification is needed in the source's rank-three toroidal discussion. Its unrestricted wording excludes $(1, 1)$, but one must also exclude the degenerate unit periods. For example,

$$\langle x, y \mid x^3, y^3, (x^{-1}y)^3, xy^{-1}x \rangle = \langle x \mid x^3 \rangle = C_3,$$

since $y = x^2$. Its three pair-difference parabolics are all the whole group; its standard coset system has one element of each type and is not thin. The usual admissible parameter domain is therefore necessary when invoking the prior toroidal classification. We make no claim here that a finite sampling proves that classification. Every period in Table 1 has norm at least three, and every Table 2 period has norm at least four; all are checked separately.

The source attributes the regular classification of locally toroidal polytopes to McMullen–Schulte, the rank-three chirality criterion to Breda D'Azevedo–Nedela, and the toroidal hypermap families to Corn–Singerman. Those classification statements remain explicit prior inputs wherever the general source recovery uses them. Our two finite tables are now verified positive examples independently of their completeness. Problem 9.2 and the unnumbered completeness conjecture remain separate questions. The old negative answers to Problems 9.3 and 9.4, including their restricted Lean scope, are acquired in the general recovery and receive no new credit here.

The source presentations, examples and positive conclusions belong to the source authors and their cited predecessors. The finite coset certificate theorem and its original implementation are reused from the accepted L003 recovery. The present contribution consists of the literal L052 applications, independent permutation and incidence checks, exact named-group maps, and the explicit corrections above. Named finite-group constructors, exact GAP presentation routines, and the named CHEVIE root model are declared computational and identification premises. They are not hidden inside a claim of Lean verification or new classification.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Highly symmetric hypertopes*, Aequationes Math. 90 (2016), 1045–1067; accepted author manuscript, fixed 22-page edition.

[2] *Exact rotation presentations and coset certificates* and *Literal local periods and actual marked quotients*, accepted common proofs in the complete L003 recovery, Sections 00–01 of its shared supplement; exact editions and acceptance records acquired.

[3] *Chamber propagation and two regular orbits*, accepted Topic Chamber common manuscript, version 2.

[4] G. Malle and J. Michel, CHEVIE data for G_{29}, `src/tbl/cmplxg29.jl`; four literal root vectors, retrieved from the authors' software repository. See J. Michel, J. Algebra 435 (2015), 308–336, for the package.